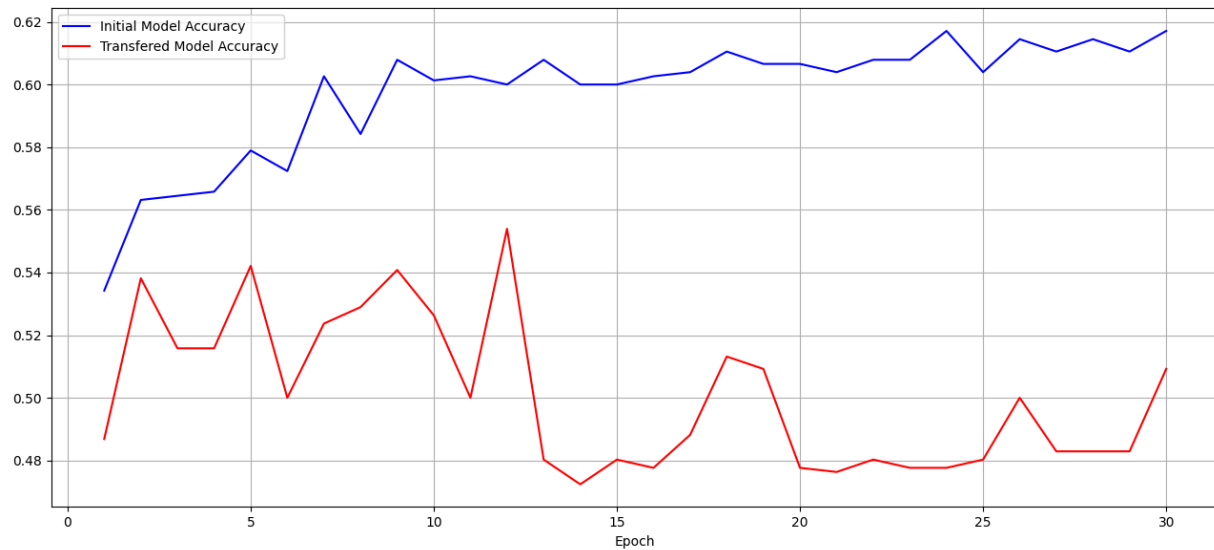


Transfer Learning Report

Far Transfer from Facial Recognition to Dog vs Cat recognition



Final Classification Accuracy on Test Set:

With transfer

* Evaluating transfered_model

4/4 [=====] - 0s 57ms/step - loss: 0.6318 - accuracy: 0.6343

Without transfer

* Evaluating random_model

4/4 [=====] - 0s 58ms/step - loss: 0.6869 - accuracy: 0.5174

Facial Recognition Classes images

Happy: 

Neutral: 

Surprised: 

Dog and Cat Classes images



Model Summaries

With Transfer:

* Training transfered_model for 30 epochs

Model: "model"

Layer (type)	Output Shape	Param #
rescaling_input (InputLayer)	[(None, 150, 150, 3)]	0
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 147, 147, 16)	784
max_pooling2d (MaxPooling2D)	(None, 36, 36, 16)	0
dropout (Dropout)	(None, 36, 36, 16)	0
conv2d_1 (Conv2D)	(None, 33, 33, 32)	8224
max_pooling2d_1 (MaxPooling2D)	(None, 11, 11, 32)	0
dropout_1 (Dropout)	(None, 11, 11, 32)	0
conv2d_2 (Conv2D)	(None, 8, 8, 48)	24624

max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 48)	0
dropout_2 (Dropout)	(None, 4, 4, 48)	0
flatten (Flatten)	(None, 768)	0
dense (Dense)	(None, 64)	49216
dropout_3 (Dropout)	(None, 64)	0
transfer_dense (Dense)	(None, 32)	2080
transfer_output (Dense)	(None, 2)	66

Total params: 84,994

Trainable params: 2,146

Non-trainable params: 82,848

Without transfer:

* Training random_model for 30 epochs

Model: "model_1"

Layer (type)	Output Shape	Param #
rescaling_input (InputLayer)	[(None, 150, 150, 3)]	0
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 147, 147, 16)	784
max_pooling2d (MaxPooling2D)	(None, 36, 36, 16)	0
dropout (Dropout)	(None, 36, 36, 16)	0
conv2d_1 (Conv2D)	(None, 33, 33, 32)	8224
max_pooling2d_1 (MaxPooling2D)	(None, 11, 11, 32)	0
dropout_1 (Dropout)	(None, 11, 11, 32)	0

conv2d_2 (Conv2D)	(None, 8, 8, 48)	24624
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 48)	0
dropout_2 (Dropout)	(None, 4, 4, 48)	0
flatten (Flatten)	(None, 768)	0
dense (Dense)	(None, 64)	49216
dropout_3 (Dropout)	(None, 64)	0
random_dense (Dense)	(None, 32)	2080
random_output (Dense)	(None, 2)	66
=====		

Total params: 84,994

Trainable params: 2,146

Non-trainable params: 82,848
